The Mixed Income Neighborhoods Overlay and Residential Development in Philadelphia

Descriptive Trends and Causal Evidence

Prepared on behalf of
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City of Philadelphia

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Executive Summary

Philadelphia's Mixed Income Neighborhoods (MIN) overlay requires residential projects of 10 or more units to set aside 20% of these homes as affordable to households earning up to 40% of the area median income. This report evaluates the overlay's effect on residential development using two complementary frames: a descriptive trend analysis of permit volume, project size, and dwelling-type composition inside the overlay relative to the rest of the city, followed by a quasi-experimental difference-in-differences (DiD) analysis that tests whether the observed trends can be causally attributed to the policy.

The descriptive analysis uncovered three patterns inside the overlay after 2022. The ratio of new commercial-mixed-use (CMX) construction permits inside versus outside the MIN reversed from roughly 0.66 to 1.99; the share of projects in the 5–9-unit band just below the inclusionary threshold roughly doubled; and multi-family construction grew 7.3 percentage points more than in the rest of the city. Read in isolation, these patterns suggest the overlay meaningfully reshaped development.

The causal analysis shows that most of these signals reflect citywide forces rather than the overlay itself. Once MIN census tracts are compared to a matched set of non-MIN tracts that resemble them on income, tenure, vacancy, and pre-policy permitting, the apparent effects on permitting volume, project size, and threshold avoidance disappear and are statistically indistinguishable from zero. The dramatic raw slowdown is better explained by the 2021 tax-abatement redesign and the post-2022 rise in financing and construction costs than by the inclusionary requirement. The one descriptive signal that survives—multi-family construction channeling into MIN corridors—is not formally tested by the DiD and remains the clearest lead for future causal work.

In its first four years, the MIN overlay shows no measurable evidence of suppressing development, pushing back against developer concerns. These findings indicate the requirement is small enough not to deter development, or it is too early to detect an effect given the lag between entitlement and permitting. The report recommends maintaining the overlay under the current design, and re-evaluating the policy on a longer horizon as more post-2022 data accumulates.

1. Introduction

This report evaluates how Philadelphia's Mixed Income Neighborhoods (MIN) zoning overlay has affected residential development using two complementary analyses. The first is a descriptive trend analysis that maps changes in permit volume, project size, and dwelling-type composition inside the overlay relative to the rest of the city. A quasi-experimental difference-in-differences (DiD) study follows, asking whether those changes can be causally attributed to the overlay once a credible comparison group is constructed. Qualitative case studies of representative completed developments are also provided for additional context. Before diving into this analysis, it is important to establish the policy context and economic logic that motivate the empirical tests, as well as the confounding changes to the development environment that complicate any simple before-and-after comparison.

1.1 The Role of Zoning Policy

Zoning regulations are the primary policy tool municipal governments use to control the types, shapes, and uses of buildings across neighborhoods. Initially conceived in the early twentieth century to limit the impact of hazardous industrial factories on growing residential neighborhoods, zoning has evolved to encompass all forms of land use within a city[1]. In Philadelphia, City Council oversees approval of zoning codes following recommendations from neighborhood residents and city staff to meet the evolving needs of the community and the city overall.

All developments must submit a proposal showing how they abide by the zoning code. Once the city determines a project meets all the zoning requirements for a site, it is considered a "by-right" development and can advance to the next stage of review. When a development falls outside the requirements of the code, such as building an apartment building on a street zoned for rowhomes, the developer must request a zoning variance and go before the Zoning Board of Adjustment (ZBA). These public hearings weigh community input, environmental impacts, and traffic effects. Because the ZBA process can add months to a project's timeline, many developers adapt their projects to avoid the variance process which can become costly as financing costs

[1] William A. Fischel, "An Economic History of Zoning and a Cure for Its Exclusionary Effects.," *Urban Studies (Routledge)* 41, no. 2 (February 2004): 371–340, 12505591, https://doi.org/10.1080/0042098032000165271.

mount while a development is stalled. This motivation is most prevalent when there is a strict cutoff associated with the decision.

1.2 Inclusionary Zoning and The Mixed Income Neighborhoods Overlay

Housing affordability is an increasingly salient policy issue across U.S. cities. Pew's 2025 State of the City for Philadelphia reported that 51.7% of Philadelphia households were rent-burdened, spending more than 30% of income on housing, and the city's 20.3% poverty rate places it among the highest in the northeast region.[2] While the city has improved from the high concentrations of poverty seen in the 1990s and early 2000s[3], city officials now face the dual mandate of promoting growth while preserving affordability for existing residents.

Considering these challenges, policymakers have promoted inclusionary zoning policies as a mechanism for increasing affordable housing supplies. There are two main designs for inclusionary zoning policy. Opt-in zoning bonuses provide projects with relaxed requirements, such as additional building height, in response to reserving a portion of the finished units for rent at below market rates. This contrasts with mandatory overlays, which lay out hard requirements on strategic neighborhoods for new developments for affordable housing provision[4].

The Mixed Income Neighborhoods zoning overlay is an example of the latter, enacted by Philadelphia City Council in January 2022 across six areas of the city. The ordinance requires new construction or substantial rehabilitation projects with 10 or more residential units to reserve 20% of the units at a price affordable to residents earning up to 40% of the area median income (AMI)[5]. It is intended to preserve affordability on the periphery of faster-growing neighborhoods, as shown in Figure 1.

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[2] "Philadelphia 2025: The State of The City," accessed March 30, 2026, https://pewtrsts.org/4cpRxQu.
[3] "How Philadelphia's Housing Market Has Changed Since 2000," accessed March 30, 2026, https://pew.org/3S9SJys.
[4] Ruoniu Wang and Xinyu Fu, "Examining the Effects of Policy Design on Affordable Unit Production Under Inclusionary Zoning Policies," *Journal of the American Planning Association* 88, no. 4 (October 2022): 550–64, https://doi.org/10.1080/01944363.2022.2027263.
[5] "§ 14-533. /MIN, Mixed Income Neighborhoods Overlay District.," American Legal Publishing, accessed March 23, 2026, https://codelibrary.amlegal.com/codes/philadelphia/latest/philadelphia_pa/0-0-0-291906.

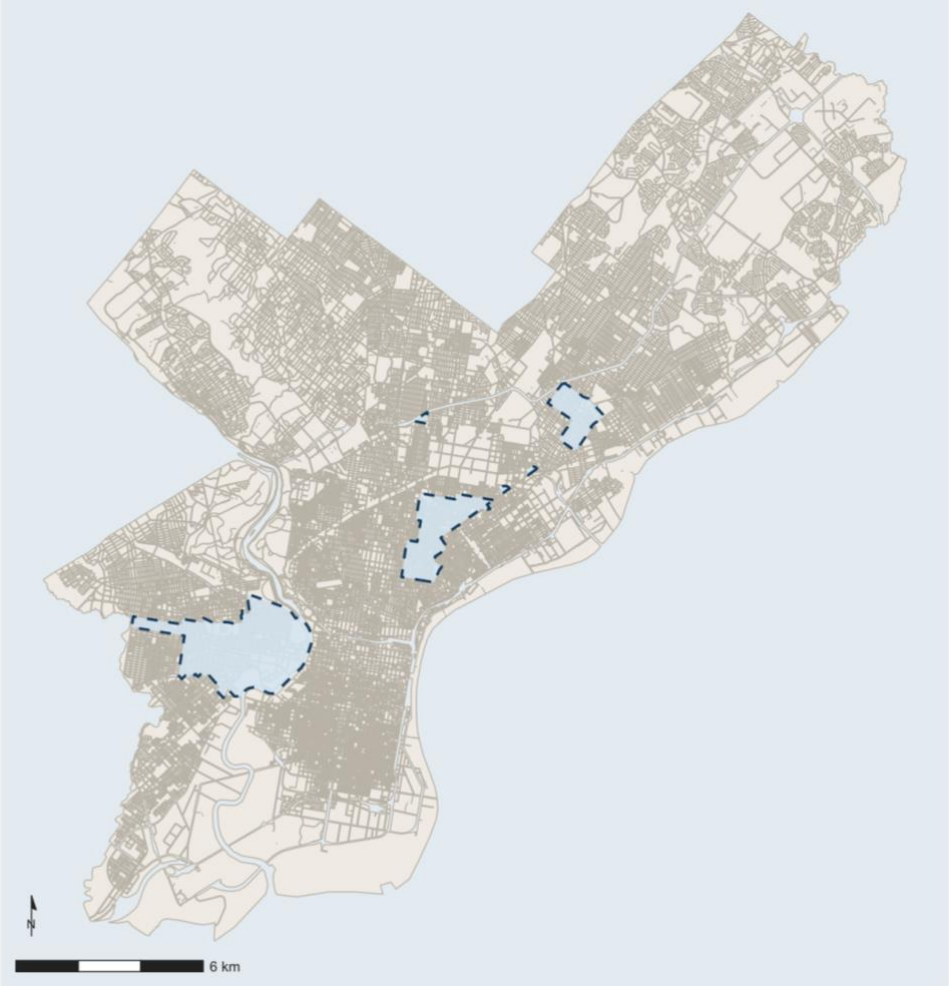


Figure 1: Boundaries of MIN overlay

1.3 Economic Theory

Economic theory treats mandatory inclusionary zoning as an implicit tax on new development[6]. As shown in the table below, the difference in Philadelphia between rent affordable to 40% area median income and 40th percentile market rate rents is substantial and would require any development to see a large reduction in profitability, and ultimately access to financing as investors opt to invest in areas with higher returns.

[6] Ruoniu (Vince) Wang, Wei Kang, and Xinyu Fu, "Do Inclusionary Zoning Policies Affect Local Housing Markets? An Empirical Study in the United States," *Cities* 158 (March 2025): 105736, https://doi.org/10.1016/j.cities.2025.105736; Vinit Mukhija et al., "Can Inclusionary Zoning Be an Effective and Efficient Housing Policy? Evidence from Los Angeles and Orange Counties," *Journal of Urban Affairs* 32, no. 2 (May 2010): 229–52, https://doi.org/10.1111/j.1467-9906.2010.00495.x; Edward Glaeser and Joseph Gyourko, "The Economic Implications of Housing Supply," *Journal of Economic Perspectives* 32, no. 1 (February 2018): 3–30, https://doi.org/10.1257/jep.32.1.3.

Table 1: Affordable Rental Prices and Market Rate prices in Philadelphia		
Unit Type	Rent affordable for 40% AMI	HUD Fair Market Rent[7]
Studio	$803	$1,303
1 Bedroom	$860	$1,451
2 Bedroom	$1,032	$1,737
3 Bedroom	$1,193	$2,100

Accordingly, the Mixed Income Neighborhoods policy has few supporters in among Philadelphia Developers[8]. One developer active in West Philadelphia willing to answer questions in relation to this project described the policy as the "death knell" for Philadelphia real estate development.

Supply-side theory predicts this imbalance will produce some combination of three developer responses: (1) Reducing total project starts, (2) Shifting toward projects that fall below the 10-unit threshold, or (3) Increasing the size of qualifying projects to spread compliance costs across additional market-rate units. Each of these mechanisms leaves a distinct signature in permit data, and together they motivate the three research questions and the empirical tests that follow.

1.4 Confounding Policy and Macroeconomic Shocks

Two main confounding shocks complicate any direct pre- versus post-2022 comparison. First, Philadelphia's tax abatement was reformed in 2021, prompting a citywide permit rush in 2020–2021 as developers filed in advance of the change[9]. In addition, during the post-pandemic inflationary period the Federal Reserve began an interest-rate hiking cycle, raising development financing costs while overall construction costs ballooned[10]. Both shocks affected MIN and non-MIN areas simultaneously. However, with their timing coming immediately before and around

[7] "HUD Fair Market Rents (FMRs) By State, County, and Metropolitan Areas | HUD USER," accessed July 2, 2026, https://www.huduser.gov/portal/datasets/fmr.html.
[8] Taylor Allen, "Quiñones-Sánchez Revives Push for Mandated Affordability in Some Gentrifying Areas," News, *WHYY*, June 24, 2021, https://whyy.org/articles/philly-council-to-mandate-affordability-in-certain-gentrifying-areas-under-new-bill/.
[9] Peter F. Kelsen Klein (LAW '04) Adam E. Laver, Jared N., "Philadelphia City Council Enacts Sweeping Changes to the Tax Abatement Program," *The Temple 10-Q*, February 27, 2021, https://law.temple.edu/10q/philadelphia-city-council-enacts-sweeping-changes-to-the-tax-abatement-program/.
[10] Dean Dalvit, "The Impact of Federal Interest Rates on Development Costs," *EVstudio*, January 31, 2025, https://evstudio.com/the-impact-of-federal-interest-rates-on-development-costs/.

the overlay's effective date, a standard before-after comparison would mis-attribute their effects to the policy. Disentangling these changes is the central methodological challenge this report addresses.

1.5 Case Studies of Completed MIN and Comparison Developments

To provide context for the permit-level patterns with on-the-ground detail, this section profiles developments that had reached occupancy under the overlay as of early 2026, alongside two comparison projects built outside it. These qualitative cases are illustrative rather than representative, providing examples of unit counts, zoning context, and timelines that the quantitative analysis examines in aggregate.

1.5.a MIN Case Studies

To date, three projects have reached occupancy while meeting all the requirements of the MIN policy: 1611–1613 Foulkrod Street, 1615 Foulkrod Street, and 300–310 W. Norris Street. These projects span from just above the MIN threshold at 11 units to far exceeding it at 33 units, consistent with the trend seen among unfinished projects within the MIN areas.



Figure 2: Design plans for 1611 Foulkrod Street

1611–1613 Foulkrod Street. This is primarily a multifamily development with 11 units, 3 of which are affordable, plus a street-facing commercial space. The building incorporates an existing abandoned structure that appeared to house both a commercial and a residential unit, and is consistent with the RM-1 base zoning. With a total lot size of 4,790 sq ft and 8,960 sq ft of improved space, it is the smallest MIN-compliant project and sits just above the 10-unit cutoff. The project received zoning and construction approvals in January 2024 and began occupancy in March 2025. One unit is currently available, indicating nearly full lease-up.

1615 Foulkrod Street. Also an adaptive-reuse and overbuild project, and directly adjacent to the development above, this is a distinct project from a different developer. Now marketed as the Yarn Lofts, the former industrial building was in disrepair prior to 2023. The 23-unit building provides 5 affordable units under the MIN, with an average unit size of roughly 704 sq ft across one-bedroom, two-bedroom, and bi-level floorplans. Although the parcel is zoned RM-1, the existing industrial building is grandfathered in. The project received zoning approval in March 2023, with construction beginning in May 2023 and occupancy starting in September 2024. The building encompasses 16,200 sq ft on a 14,655.5 sq ft site, including streetscaping and surface parking for residents. It is currently fully leased, with no units available.

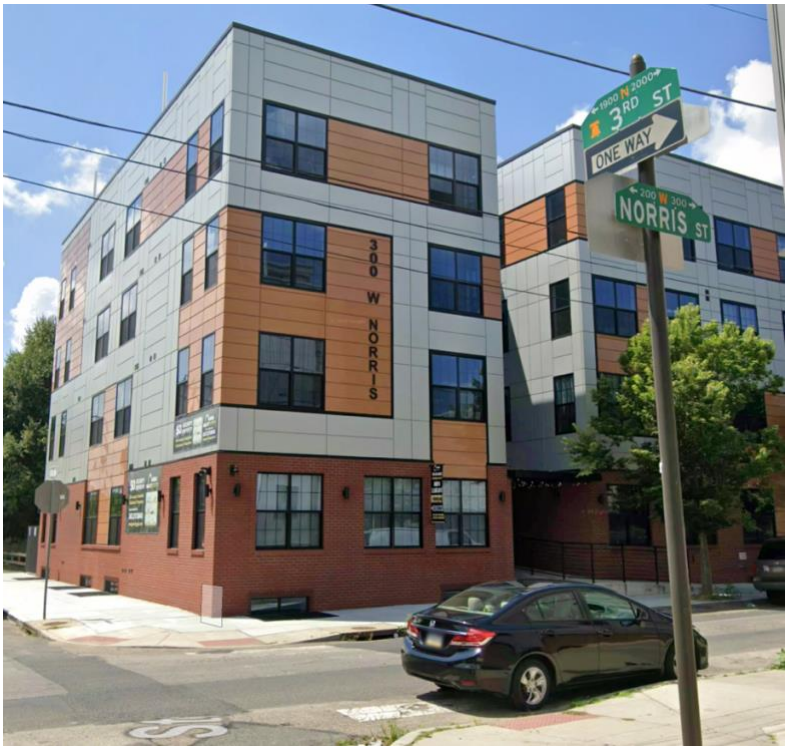


Figure 3: 300 W Norris Street Development

300–310 W. Norris Street. Initially six vacant parcels, this property was developed into 33 apartments, supplementing 24 market-rate units with 10 affordable units. In addition to providing the 20% affordable set-aside required under the MIN, the project takes advantage of the Mixed Income Housing Bonus (MIHB) to provide an additional three affordable units. The building contains 21,800 sq ft of space on a 6,839 sq ft lot, with an average unit size of roughly 660 sq ft. The project began the zoning process in February 2023, with construction beginning the following year in February 2024 and passing final inspection in May 2025. As of March 2025, the building is fully leased, with no units available for rent.

1.5.b Non-MIN Case Studies

Two projects, 4550 Mitchell Street in Roxborough and 1600 Point Breeze Avenue in Point Breeze, were selected as representative case studies of multifamily development outside the MIN overlay.

4550 Mitchell Street. Branded as The Aster, this mixed-use development of primarily residential units in Roxborough with floor-level retail, contains 38 units, 4 of which are designated affordable to take advantage of the Mixed Income Housing Bonus. On the 10,842 sq ft CMX-2-zoned lot, a maximum of 22 units would be allowable without the MIHB and the green-roof bonus. The project received zoning approval in February 2022 and construction approval in March 2022, with occupancy beginning in January 2025. The building is nearly full, with one one-bedroom unit available.

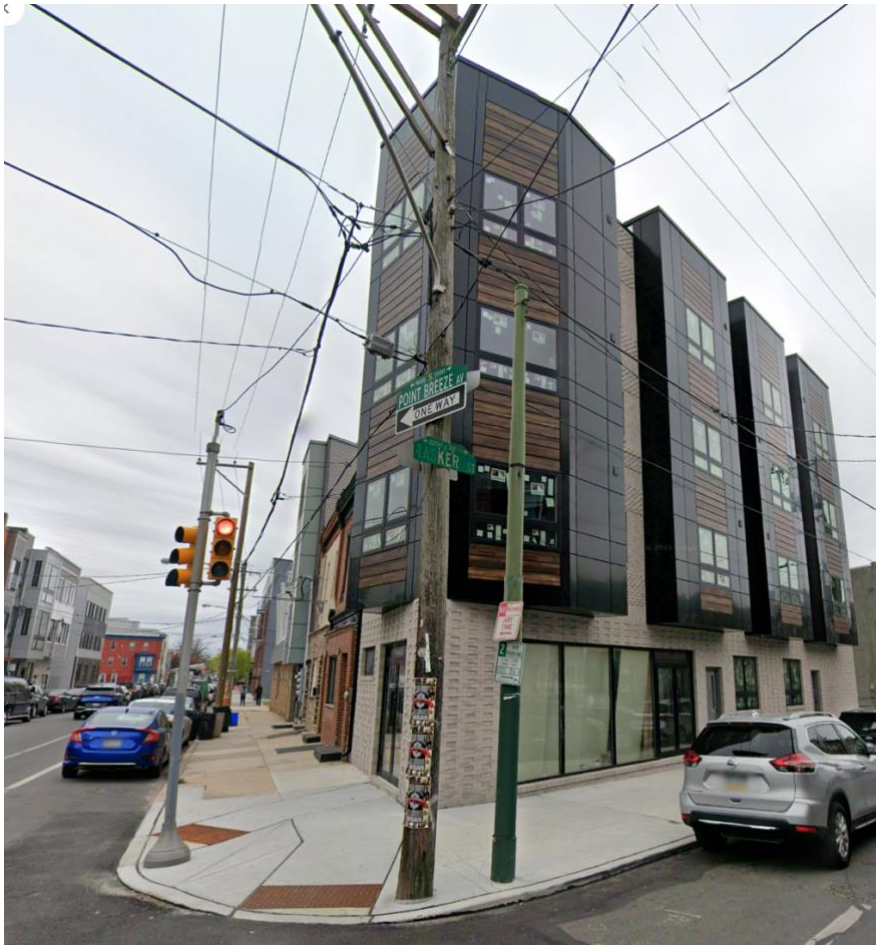


Figure 4: 1600 Point Breeze Avenue

1600 Point Breeze Avenue. This South Philadelphia corner building formerly hosted a bar and apartments, which were demolished sometime between 2011 and 2014 according to Google Streetview. The current project brings back a similar use case of ground-level commercial with six residential units. It does not use any zoning bonuses and is entirely market-rate. Given the smaller lot size of 1,586 sq ft, the developer was required to lower the building height from 52.5 ft to 39 ft, slightly above the allowable limit, while maintaining the same unit count. The development received final zoning approval in December 2021, with construction beginning in June 2022. As of February 2026, however, the project has stalled with no timeline for final occupancy approval, though one year remains on the construction permit. Together, these two projects illustrate the risk involved in residential development and the opportunities available to maximize leasable space.

With this background information contextualizing the Mixed Income Neighborhoods program within Philadelphia and the broader inclusionary zoning policy domain, we turn to evaluating the performance of the policy over the first four years in effect.

1.6 Research Questions and Analytical Strategy

This report asks whether the MIN overlay changed residential development along the three margins that economic theory predicts, and whether any such change can be distinguished from citywide trends. It organizes the evidence around three research questions:

(1) Did the overlay depress permitting volume?

(2) Did developers shrink projects or bunch them below the 10-unit threshold?

(3) Did the composition of new projects change?

Each question is examined in two stages, first focusing on a raw comparison of the trends, followed by a causal pass (the difference-in-differences design) testing whether the trend can be attributed specifically to the overlay.

2. Data and Methods

This section describes the shared data foundation, and the two analytical approaches built on it: the descriptive trend analysis, and the causal difference-in-differences design. Both analyses draw on the same underlying permit and zoning records, which is what allows their findings to be read against one another in Section 3.

2.1 Data Sources

Permit data are drawn from the Philadelphia L&I Construction Permit Records, accessed through the city's API. Parcel-level zoning information comes from the Philadelphia City Planning Commission's Zoning Base Districts and Zoning Overlays shapefiles, and parcel geometries from the Philadelphia Department of Records parcels file. All datasets are provided through the OpenDataPhilly portal. A spatial join links each permit to the parcel containing its address, attaching base zoning and overlay codes. For the causal analysis, permits are additionally geocoded and assigned to 2020 census tracts.

2.2 Data Cleaning and Processing

MIN-overlay permits are identified as records on parcels whose overlay code field contains the substring "/MIN". Zoning overlays are the main vehicle for geography-specific policies, and a single parcel may carry multiple overlays. This filter yields 12,850 permits on 6,569 unique parcels issued between 2007 and 2026. Permits are split into pre-2022 (issued before January 1, 2022) and post-2022 (on or after that date). Unit-count and dwelling-type data are derived from the free-text description field, parsed with a series of regular-expression filters; these fields are available for approximately 20.5% and 39% of MIN permits, respectively. This incomplete coverage is a recurring caveat for the size- and composition-based findings and is revisited in Section 5.

2.3 Descriptive Trend Analysis

The descriptive analysis has three components, each aligned with one of the research questions in Section 1.6. First, it assesses geographic patterns over time by mapping MIN-specific permit volume against policy boundaries and comparing volume pre- and post-2022. Second, it bins the unit-count distribution within the overlay to gauge the extent of threshold avoidance relative to the alternative of spreading marginal compliance costs over additional units. Third, it evaluates compositional change in dwelling types inside the MIN relative to greater Philadelphia, pre- and post-2022. Because it uses the full permit record, the descriptive analysis offers broad coverage and surfaces candidate signals. However, as Section 2.4 explains, it cannot by itself separate the overlay's effect from citywide market forces.

2.4 Difference-in-Differences Analysis

While the trend analysis is helpful in uncovering potential relationships, that cannot alone provide the basis for determining whether the same patterns have appeared without the overlay. Because the 2021 abatement redesign and the post-2022 macroeconomic cycle influenced development citywide, a causal estimate requires a comparison group of non-MIN areas that behaves as the MIN areas would have absent the policy. The difference-in-differences design constructs that counterfactual in the following three steps.

2.4.a Treatment Definition and Panel

A census tract is treated if at least 50% of its land area lies within the /MIN overlay, yielding 31 treated tracts. The 50% cutoff ensures that all tracts with a majority exposure are captured in the treatment group. At lower cutoffs, a treated tract is mostly land not covered by the Mixed Income Neighborhoods policy, and sensitivity is tested in the robustness checks summarized in Section 3.1.b. The data are organized as a tract-by-month panel of permit counts for all tracts, alongside a permit-level file for the units-per-permit outcome. Each outcome is reported for two families side by side: a CMX-only family that isolates the commercial-mixed-use zones the overlay principally impacts, and an all-parcel family that captures any spillover or substitution into other zoning types.

2.4.b Comparison Group Construction

MIN census tracts are not representative of the average Philadelphia tract: they are lower income, have higher renter and vacancy shares, and showed different pre-2022 permitting than the city overall. Comparing them to all other tracts would confound the overlay with these baseline differences. Therefore, it is important to select only the non-MIN tracts that resemble MIN tracts across measurable criteria excluding the overlay. Matching is performed across 15 metrics detailed in Table 2 below.

Table 2. Matching Variables for Comparison Group

Renter Share	Median household income	Median age
Average household size	CMX land share	Housing Density
Race/ethnicity shares (4 variables)	Vacancy Rate	Pre-2022 permitting metrics (4 variables)

Restricting only to tracts that contain some CMX land, we construct the comparison group using a propensity distance matching model. With only 29 treated tracts, a geometric distance across 15 variables cannot balance each one simultaneously, trading imbalance from one variable to another. The propensity score instead collapses the 15 variables into a single dimension measuring likelihood of treatment. After matching, the largest standardized difference on any covariate is 0.219, and all 29 treated tracts are retained.

2.4.c Estimation and Inference

Effects are estimated with a two-way fixed-effects event study and a pooled difference-in-differences analysis. Tract fixed effects absorb geographic differences between tracts, while month fixed effects absorb chronological variations impacting the entire city such as the abatement redesign, interest rates, and seasonality. What remains is the change in treated tracts after January 2022 relative to the matched controls' change, thought of as the difference within the difference. Counts are modeled with Poisson regression, which is robust to the many zero-permit tract-months in the panel[11]. Because the abatement came into effect for all areas at the same date, the standard two-way fixed effects design is appropriate.

Inference. A causal estimate is only as credible as its underlying uncertainty, and the clustered standard error is not necessarily trustworthy in this setting. Each pooled estimate therefore reports three errors:

(1) Two-way clustered error at the tract and month level as the baseline.

(2) Conley spatial error, because neighboring tracts share spillover shocks such as a hot development corridor impacting adjoining tracts[12]. The requisite Moran's diagnostic confirms strong spatial correlation.

(3) CR2 small-sample correction, which is the appropriate fix when only 30–50 treated clusters make ordinary clustered errors prone to over-rejection[13].

Finally, a formal parallel trends check tests whether treated and control tracts were already diverging in the 12 months before the overlay. The window of 2015-2019 is tested rather than all 15 pre-years, because the early CMX series is too sparse to be informative.

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[11] Jeffrey M. Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, in *Second Edition. Cambridge and London: MIT Press, 2010, Pp. Xxvii, 1064* (Second edition. Cambridge and London: MIT Press, 2010), xxvii, 1064, https://www.proquest.com/docview/874880210?https://hdl.library.upenn.edu/1017/6893&pq-origsite=summon&sourcetype=Books.

[12] T. G. Conley, "GMM Estimation with Cross Sectional Dependence," *Journal of Econometrics* 92, no. 1 (September 1999): 1–45, https://doi.org/10.1016/S0304-4076(98)00084-0.

[13] James E. Pustejovsky and Elizabeth Tipton, "Small Sample Methods for Cluster-Robust Variance Estimation and Hypothesis Testing in Fixed Effects Models," *Journal of Business & Economic Statistics* 36, no. 4 (October 2018): 672–83, https://doi.org/10.1080/07350015.2016.1247004.

3. Findings

The findings are organized around the three research questions from Section 1.6. For each question, the descriptive trend is presented first, followed by its difference-in-differences test, and then a short synthesis of what the two readings jointly imply. Table 3 reports the full set of pooled DiD estimates referenced throughout this section. Overall, descriptive signals which look striking in isolation tend to shrink toward zero once the matched comparison group is evaluated.

Table 3. Pooled Difference-in-differences estimates, by outcome variable and population

Outcome (family)	Estimate	Plain reading	p (clustered / Conley)
Units per permit (CMX)	+0.15 log	~+16%, n.s.	0.69 / 0.57
Permit count (CMX)	+0.03	~+3%, n.s.	0.95 / 0.96
5–9-unit permits (CMX)	+0.58	n.s.	0.44 / 0.57
Units per permit (all)	+0.24 log	**~+27%**	0.037 / <0.001
Permit count (all)	−0.21	~−19%, n.s.	0.54 / 0.50
5–9-unit permits (all)	+0.16	n.s.	0.70 / 0.74

3.1 Did the MIN Depress Permitting Volume?

3.1.a Descriptive Evidence

Of the 12,850 MIN-overlay and 122,272 non-MIN permits, 81% were issued before 2022 and 19% after in both groups. On a simple read, both MIN and non-MIN areas slowed in tandem. The more pronounced signal is in the commercial-mixed-use (CMX) corridors which the overlay primarily regulates. Restricting to CMX new-construction permits holds base zoning fixed and removes one source of attenuation between the two groups. As Table 4 shows, the inside-to-outside CMX permit ratio reverses over the study period. The 2015–2019 baseline is roughly 0.66, growing to 1.99 by 2025. In other words, MIN corridors shifted from producing about two-thirds as many CMX permits as non-MIN areas, to MIN-area producing nearly twice that of non-MIN areas. Year-over-year changes confirm the asymmetry: from 2021 to 2022, CMX new construction fell 18.8% inside the overlay versus 83.4% outside, and from 2022 to 2025 the inside-MIN series moved only modestly while the citywide series was far more volatile as shown in Figure 6.

Figure 5: Event Study Plot by Outcome variable

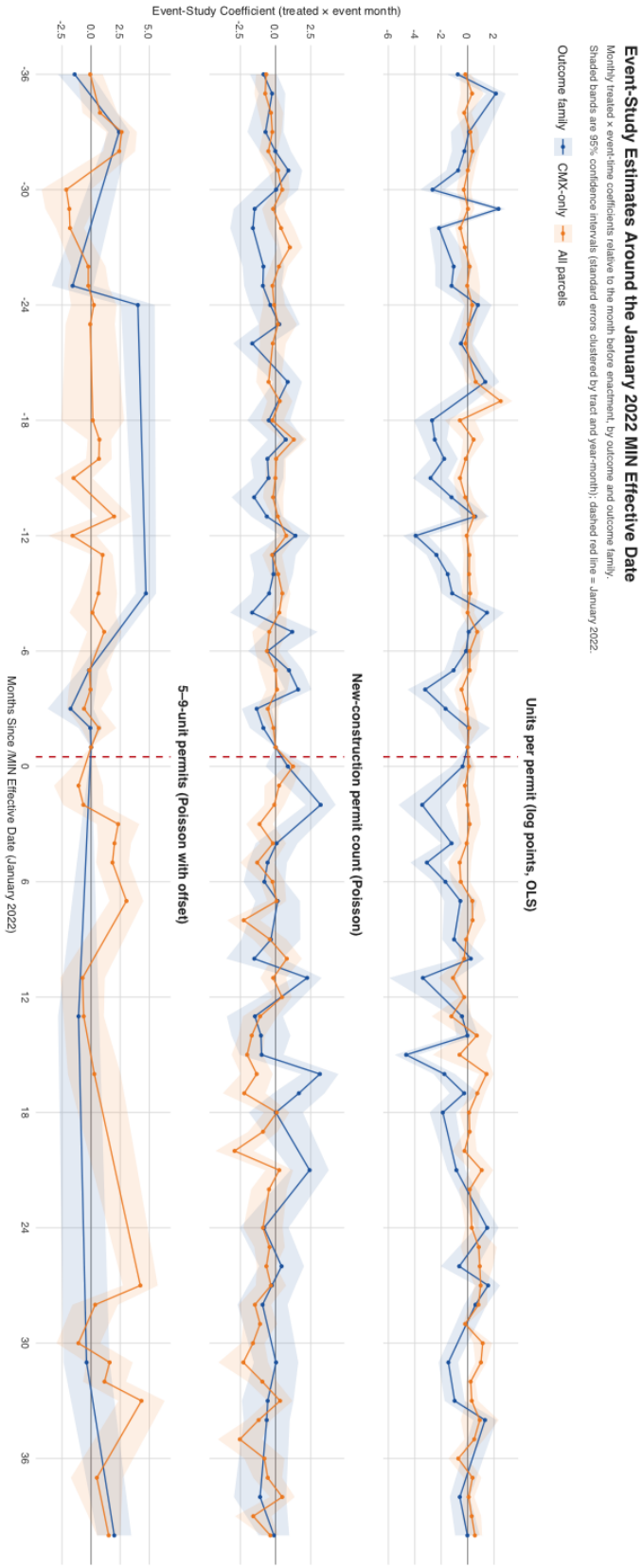


Table 4. CMX new-construction permits, MIN overlay vs. non-MIN areas.

Period	Inside MIN (CMX permits/yr)	Outside MIN (CMX permits/yr)
2015–2019 baseline	~202	~307
2021 peak (abatement-driven)	303 (+50% vs. baseline)	469 (+53% vs. baseline)
2025	231 (+14% vs. baseline)	116 (−62% vs. baseline)
Inside/outside ratio (2015–2019)	0.66	—
Inside/outside ratio (2025)	1.99	—

Taken at face value, this reversal could suggest that demand inside MIN corridors is robust enough to be unaffected by the requirement. But it is equally consistent with differential exposure of MIN and non-MIN CMX corridors to the post-2022 financing environment and inventory glut. Distinguishing between these readings is exactly what the causal test is designed to do.

3.1.b Causal Results

Once the comparison group is matched, the volume effect disappears. The CMX permit-count estimate is +0.03 (about +3%) and statistically insignificant under both standard errors ($p = 0.95$ clustered, 0.96 Conley). The all-parcel permit-count estimate is −0.21 (about −19%) and likewise insignificant ($p = 0.54$ / 0.50). The null is stable across robustness checks: varying the treatment threshold from 25% to 75% area overlap moves the permit-count effect only within −0.52 to +0.29 and never to significance; dropping the 2020–2021 COVID/abatement window ("donut"), using all non-MIN tracts unmatched, and shifting the effective date to 2020 or 2021 to allow for the 1–2-year permit pipeline all return null results.

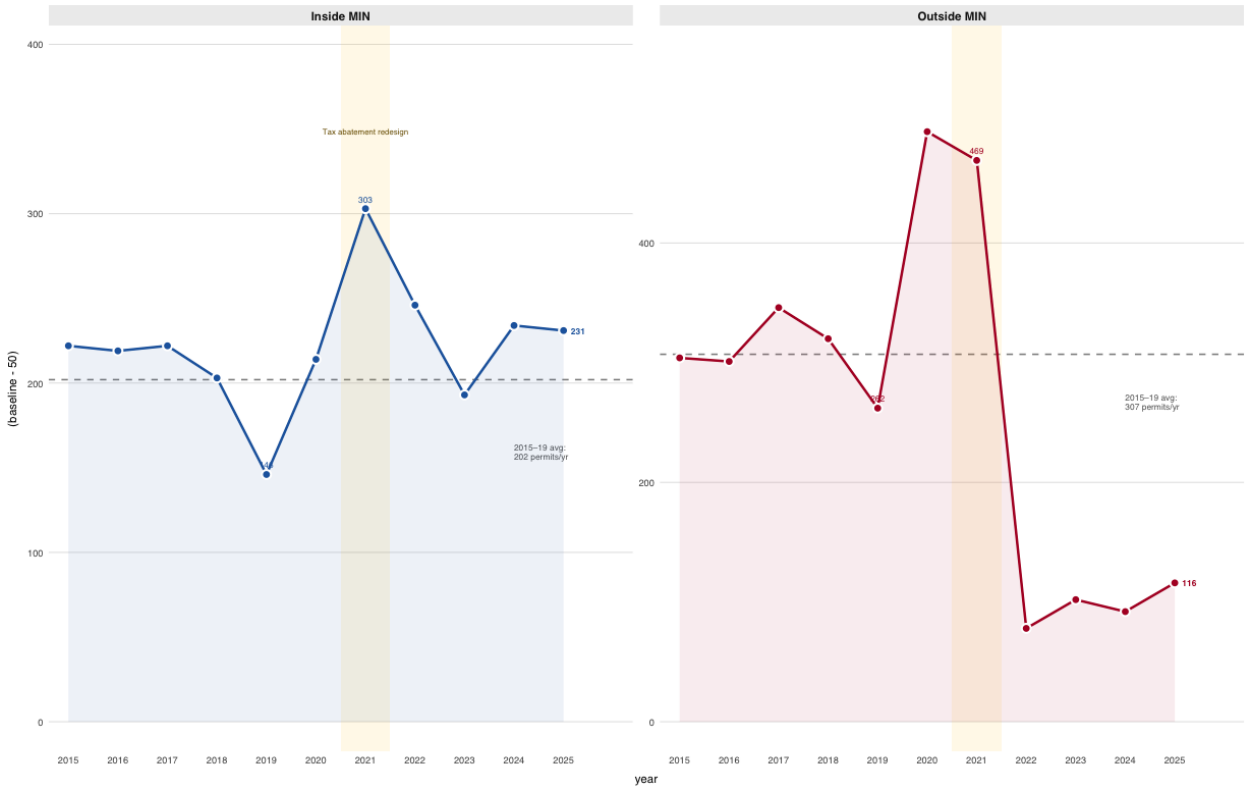
3.1.c Combined Analysis

The descriptive reversal from Table 3 is dramatic, however the causal test indicates it is largely an artifact of comparison. MIN corridors were being measured against dissimilar non-MIN areas that were more exposed to the abatement-driven boom and bust. The outside-MIN CMX series fell 62% below its baseline by 2025 not because of any impact from the overlay, but because

those areas simply had farther to fall. Once non-MIN tracts are chosen to resemble MIN tracts on income, tenure, vacancy, and pre-period permitting, the apparent volume divergence vanishes.

The answer to the first research question is therefore no: there is no evidence the overlay depressed permitting volume in its first years.

Figure 6: CMX New Construction Permits: MIN overlay vs. Outside, 2015-2025



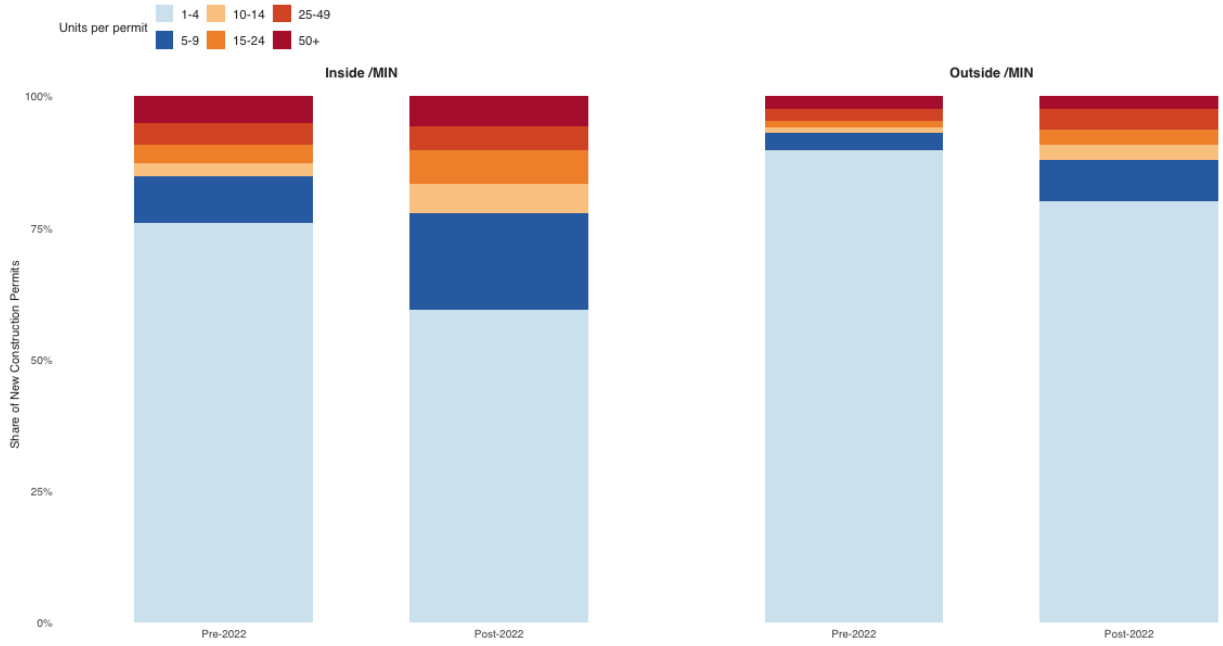
3.2 Did Developers Shrink Projects or Bunch Below the 10-Unit Threshold?

3.2.a Descriptive Evidence

Within MIN areas, average project size rose from 6.21 units pre-2022 to 16.74 units post-2022, while citywide average size grew more modestly from 3.62 to 6.74 units. At the same time, the share of MIN permits in the 5–9-unit band, just below the 10-unit inclusionary trigger, grew from 8.9% to 18.3%, roughly doubling, at the expense of the modal 1–4-unit group. On the surface these two facts point in opposite directions: a rising average is consistent with developers scaling qualifying projects up to spread compliance costs, while growth in the 5–9 band is the

signature of threshold avoidance. Because unit counts are available for only about a fifth of permits, the distribution in Table 5 is informative about direction but not precise magnitudes.

Figure 7: Unit-Count Share of New Construction Permits: MIN vs. Outside, Pre- vs Post 2022



Importantly, the 5–9-unit band also grew citywide from 3.2% to 7.9%, showing descriptive pattern is not unique to the overlay, again highlighting the need for a matched comparison.

Table 5. Unit-count distribution of MIN-overlay and greater Philadelphia permits, pre- vs. post-2022.

Unit bins	Within MIN Pre-2022	Within MIN Post-2022	Within MIN Change (pp)	Greater Philadelphia Pre-2022	Greater Philadelphia Post-2022	Greater Philadelphia Change (pp)
1–4	75.9%	59.4%	−16.5	89.8%	80.1%	−9.7
5–9	8.9%	18.3%	+9.4	3.2%	7.9%	+4.7
10–14	2.5%	5.7%	+3.2	1.1%	2.8%	+1.7
15–24	3.4%	6.3%	+2.9	1.3%	3.0%	+1.7
25–49	4.2%	4.6%	+0.3	2.3%	3.8%	+1.5
50 +	5.1%	5.7%	+0.6	2.3%	2.5%	+0.2
Total (n)	**1,062**	**175**	—	**12,096**	**1,616**	—

3.2.b Causal Test

The causal estimates do not support threshold avoidance. Interestingly, the all-parcel estimate points to more units per permit, the indicating projects are growing rather than shrinking. This estimate is statistically significant under both the clustered and Conley errors. As the only statistically significant estimate in the study, it should be read with caution. The significance is marginal under the small-sample CR2 correction ($p = 0.053$) and is not confirmed in the robustness reruns. In addition, the CMX units-per-permit estimate is insignificant and the dedicated 5–9-unit bunching outcomes are insignificant in both families.

3.2.c Combined Analysis

The descriptive 5–9-unit growth is real but mirrors a citywide shift and does not survive as a MIN-specific bunching effect once matched. The fear that inclusionary requirements would push developers to build 9-unit projects to dodge the trigger is not causally proven. The only quantitatively notable estimate moves in the opposite direction (larger projects), and even that is too fragile to lean on. The answer to the second research question is that there is no robust evidence of threshold avoidance or project shrinkage attributable to the overlay.

3.3 Did the Composition of New Projects Change?

3.3.a Descriptive Evidence

Dwelling-type composition changed markedly inside the overlay between the two periods. Among MIN permits with structure-type data, single-family construction fell from 42.8% to 5.7% (down 37.1 points), townhouse/row-home rose from 31.8% to 69.3% (+37.5 points), multi-family rose from 15.9% to 23.5% (+7.6 points), and two-family/duplex fell from 6.0% to 1.2%. Read alone, this looks like a strong response to a policy that rewards consolidating units. But similar shifts also appear citywide: among non-MIN permits, single-family fell 39.9 points and townhouse rose 43.4 points, while multi-family was essentially flat (+0.3 points). Table 5 shows how much the MIN areas shifted in excess relative to the city overall.

Table 6. Composition change, post- vs. pre-2022 (MIN minus non-MIN dwelling-type shifts)

Dwelling type	MIN change (pp)	Non-MIN change (pp)	Excess shift	MIN-specific?
Single-family	−37.1	−39.9	+2.8	No
Townhouse / row home	+37.5	+43.4	−5.9	No
Multi-family	+7.6	+0.3	**+7.3**	**Yes**
Two-family / duplex	−4.8	−2.3	−2.5	No

Declines in single-family and duplex types, in addition to the townhouse share increase are primarily citywide phenomena, not MIN-specific responses. Multi-family's +7.3-point excess shift is the sole exception and indicates that multi-family construction grew meaningfully more inside the overlay than in non-MIN areas, where it was flat. This finding is consistent with MIN areas being more popular for multifamily development, either because developers preferred to consolidate compliance into a single larger project or because the overlay's geography overlaps with areas where multi-family was already feasible.

3.3.b Causal Perspective

Given the structure of the tract-parcel data being evaluated, composition is the one research question the difference-in-differences design cannot test directly. The multi-family excess shift therefore remains a descriptive finding rather than a causally identified effect. However, it is notable as the MIN-specific finding that is loosely consistent with the all-parcel tilt toward more units per permit from Section 3.2. Compositional change, specifically increased multi-family density, is the most plausible candidate for a directly attributable MIN effect. This is the most deserving of formal causal testing as more post-2022 data becomes available over time.

4. Discussion

The two analyses are complementary by design and reading them together yields a clearer picture than either alone. The descriptive stage surfaced three potential MIN trends: a post-2022 volume slowdown and reversal in the CMX permitting ratio, a doubling of the 5–9-unit share, and a shift in dwelling-type composition. The causal stage tested the first two and found them to

be largely citywide artifacts, showing the importance of comparison-group quality. The initial contrasts of pre versus post and inside versus outside conflate the overlay with the baseline differences that distinguish the Mixed Income Neighborhoods tracts. They are lower income, have higher renter shares and vacancy rates, and have a different pre-period permitting trajectory. Reductions in development attributable to the MIN appear only when the comparison group includes dissimilar Philadelphia. Once they are adequately matched, the effects disappear. The earlier inside/outside reversal was an artifact of comparing MIN corridors to tracts that were simply more exposed to the abatement boom and the subsequent financing squeeze.

The findings validated by the matched analysis are more limited. Descriptive multi-family excess shift is the clearest MIN-specific pattern, though it is not formally tested by the DiD. The one statistically significant causal estimate, that MIN neighborhoods see roughly 27% more units per permit, is tenuous and runs counter to the threshold-avoidance story. Both are best treated as leads for future research rather than settled effects pending additional data. The overarching takeaway is that the MIN's most plausible footprint is on the composition of new projects, not on their volume or threshold positioning.

5. Limitations

Several limitations bound the strength of these conclusions, and they apply to both analytical stages. Most fundamentally, construction permits indicate intent to build, not completed construction or units delivered; a permit filed is not a home occupied. The unit-count and dwelling-type fields parsed from free text cover only about 20.5% and 39% of MIN permits, so the distributions in Tables 5 and 6 are informative about direction but not precise magnitudes, and the coverage may be biased by project type.

The causal design carries additional constraints due to low observation counts. Pre-trends are plausible but volatile for every outcome: the count outcomes are reasonably flat in the year before treatment, but the 5–9-unit and CMX units-per-permit series fail the pre-trend test, so their estimates carry less weight. Residual spatial correlation remains (Moran's $I = 0.39$ for the count outcome), which is why Conley standard errors are reported as the headline for counts. Finally, with only 29 treated tracts the study has the power to rule out large effects but not small

ones: “no significant effect” means the overlay did not produce a large, detectable shift in its first two to four years, not that the true effect is exactly zero.

6. Policy Implications and Recommendations

Having laid out the findings and limitations, this section covers how to apply these findings in practice. This section begins with a summary of the takeaways and closes with how these translate into recommendations for city planners.

6.1 Key Takeaways

For policymakers, the overall findings are reassuring. In its first four years, the MIN overlay shows no distinct effect on the volume, size, or threshold-positioning of new residential construction in the CMX corridors it covers. The concern from developers that mandatory inclusionary zoning would be disastrous for new development by suppressing starts or pushing projects below the 10-unit trigger, is not supported once the overlay is compared to a credible counterfactual. The sharp post-2022 slowdown that is visible in the raw data is better explained by the 2021 tax-abatement redesign and the post-2022 inflationary cycle than by the overlay itself.

Two interpretations are consistent with the evidence. The inclusionary requirement may simply be small enough not to deter development, or it may be too early to detect an effect given the one-to-two-year lag between entitlement and permitting. The clearest MIN-specific pattern disproportionate growth in multi-family construction inside the overlay is a desirable byproduct, channeling denser, higher-unit projects into the corridors the policy targets, though it has not yet been confirmed via causal analysis.

6.2 Recommendations

Having laid out the evidence considering the limitations, the combined descriptive and causal evidence suggest the following recommendations:

1. **Maintain the overlay as-is.** The evidence does not support repeal or rollback on development-suppression grounds; the feared chilling effect on supply does not appear in the matched estimates.

2. **Re-evaluate on a longer horizon.** Re-run the difference-in-differences analysis as more post-2022 data becomes available.

7. Conclusion

This report combined a descriptive trend analysis with a quasi-experimental difference-in-differences study to assess the early effects of Philadelphia's Mixed Income Neighborhoods overlay. The descriptive analysis documented three patterns associated with the policy's 2022 enactment: a reversal in the inside-to-outside CMX permitting ratio from 0.66 to 1.99, a near-doubling of the 5–9-unit share suggestive of threshold avoidance, and a multi-family increase 7.3 points larger than in non-MIN areas. The causal analysis showed that the first two of these dissolve once MIN corridors are compared to genuinely similar tracts: there is no detectable effect of the overlay on permitting volume, project size, or threshold positioning in its first years. What remains is the increasing share of multifamily which is the most plausible candidate for a real MIN effect. The bottom line is that the MIN overlay has not measurably suppressed development, and the dramatic slowdowns in the raw data reflect citywide forces rather than the policy.

Appendix

A. Data Sources

City of Philadelphia Atlas Records Tool. Case Study permit, satellite, and Street View imagery access. https://atlas.phila.gov/

City of Philadelphia, Department of Licenses & Inspections. Construction Permit Records. Accessed via ArcGIS FeatureServer (OpenDataPhilly). https://opendataphilly.org/datasets/licenses-and-inspections-building-and-zoning-permits/

Philadelphia City Planning Commission. Zoning Base Districts Shapefile. OpenDataPhilly. https://opendataphilly.org/datasets/zoning-base-districts/

Philadelphia City Planning Commission. Zoning Overlays Shapefile. OpenDataPhilly. https://opendataphilly.org/datasets/zoning-overlays/

Philadelphia Department of Planning and Development. Land Use Parcel Shapefile. OpenDataPhilly. https://opendataphilly.org/datasets/land-use/

B. App Link:

As a part of the deliverable package for this project, I developed an R shiny dashboard that natively connects to OpenDataPhilly APIs and pulls up-to-date permit information. This will allow Division of Development Services Staff to continuous track performance as eligible projects move through the development process.

The app is available here: https://willmccull.shinyapps.io/live_app/

C. Annotated Bibliography

City of Philadelphia. "§14-533. /MIN, Mixed Income Neighborhoods Overlay District." Philadelphia Zoning Code. American Legal Publishing. Accessed March 23, 2026. https://codelibrary.amlegal.com/codes/philadelphia/latest/philadelphia_pa/0-0-0-291906

This is the primary legal text of Philadelphia's Mixed Income Neighborhoods (MIN) Zoning Overlay, the mandatory inclusionary zoning policy enacted in January 2022. The ordinance requires new construction or substantial rehabilitation projects of ten or more residential units within designated overlay areas to reserve twenty percent of units at rents affordable to households earning up to forty percent of the area median income. The statute specifies the geographic boundaries of the six overlay sub-areas, the affordability thresholds, and the compliance mechanisms available to developers. As the governing legal framework for the policy under evaluation, this source establishes the regulatory parameters that shape the development incentives analyzed in both the qualitative trend analysis and the planned difference-in-differences study. The specific ten-unit threshold and twenty percent set-aside are the treatment conditions whose effects on permit

volume, project size, and dwelling type composition are examined throughout the capstone. The statute's provisions are valid for the City of Philadelphia and apply only within the designated overlay boundaries.

Denney, Jacob T. "Valuing Naturally Affordable Housing: Evidence from Los Angeles and Orange Counties of How Inclusionary Zoning Reduces Affordability." Master's thesis, Georgetown University, 2018. https://repository.digital.georgetown.edu/handle/10822/1050822

Denney uses American Community Survey data to estimate the effects of inclusionary zoning on affordability rates in Los Angeles and Orange Counties. The thesis argues that IZ policies, while producing designated affordable units, may inadvertently reduce overall naturally occurring affordable housing by raising development costs and constraining supply. This finding complicates the assumption that IZ policies increase affordability and is relevant to the Philadelphia capstone as a cautionary lens: even if the MIN overlay produces compliant affordable units, broader market affordability could deteriorate if supply constraints tighten. The ACS-based empirical approach offers a template for measuring affordability outcomes beyond simple unit counts, complementing the permit-level analysis in the qualitative trends document. The study's findings are most applicable to high-cost California markets with established IZ programs and may not transfer directly to Philadelphia's lower-cost, post-industrial housing market, where baseline construction costs and land values differ substantially.

Glaeser, Edward L., and Joseph Gyourko. "The Economic Implications of Housing Supply." *Journal of Economic Perspectives* 32, no. 1 (Winter 2018): 3–30. https://doi.org/10.1257/jep.32.1.3

Glaeser and Gyourko review the economics of housing supply in the United States, employing a cost-based framework that compares market prices to physical construction costs using R.S. Means Company data across more than one hundred metropolitan areas. They find that in high-cost markets, the gap between price and production cost functions as a regulatory tax, suggesting that land use regulations, not construction expenses, drive price inflation. In lower-cost markets, housing supply is highly elastic and prices track production costs closely. This framework provides the theoretical foundation for understanding the MIN overlay as an implicit development tax, a characterization used throughout the qualitative analysis. The authors' elasticity argument is directly relevant to the DiD design: if Philadelphia's housing supply is relatively elastic, the MIN's cost burden may suppress development more visibly than in supply-constrained cities. The findings are most valid for U.S. metropolitan areas and are strongest for single-family construction markets where the cost comparison methodology is most precise.

Hamilton, Emily. "Inclusionary Zoning and Housing Market Outcomes." *Cityscape* 23, no. 1 (2021): 161–194. https://www.jstor.org/stable/26999944

Hamilton constructs a novel dataset on inclusionary zoning programs in the Baltimore-Washington metropolitan region and applies a difference-in-differences design to estimate effects on market-rate house prices and building permits. She creates a policy stringency index capturing the set-aside percentage, income targeting depth, and offsetting subsidies such as density bonuses. The study finds evidence that IZ increases market-rate house prices but no statistically significant reduction in housing permits. This methodology is closely aligned with the planned DiD analysis for the Philadelphia capstone: Hamilton's use of permit data as the dependent variable and her construction of a policy index parallel the approach needed to isolate the MIN's effect from confounding shocks like the tax abatement redesign and interest rate cycle. Her null finding on permits suggests that supply effects may manifest through project composition rather than volume is consistent with the threshold avoidance patterns documented in the qualitative analysis. Findings are most applicable to mid-Atlantic metropolitan areas with comparable housing market dynamics.

Kontokosta, Constantine E. "Do Inclusionary Zoning Policies Equitably Disperse Affordable Housing? A Comparative Spatial Analysis." *Journal of Housing and the Built Environment* 30, no. 4 (November 2015): 569–590. https://doi.org/10.1007/s10901-014-9430-5

Kontokosta examines the spatial distribution of over 11,000 inclusionary zoning units built between 1980 and 2000 in Montgomery County, Maryland, and Suffolk County, New York. Using a geo-coded property-level database and comparative spatial analysis, the study finds evidence of spatial clustering of IZ units despite policies designed to promote dispersal, with clustering patterns varying based on whether housing and land use authority is regional or local. This spatial methodology parallels the GIS-based permit mapping in the qualitative trends analysis, which documents geographic concentration of MIN permits in North and West Philadelphia. Kontokosta's finding that institutional framework shapes IZ outcomes is relevant to evaluating Philadelphia's overlay, where City Council controls both zoning and overlay boundaries. The study's validity is strongest for suburban jurisdictions with longstanding IZ programs and may not generalize to urban contexts like Philadelphia where overlay boundaries were recently established and cover diverse neighborhood types.

Luka, Nik, Richard G. Shearmur, Leila Ghaffari, A. J. Bimm, and Josh Medicoff. "Inside Montréal's Inclusionary Zoning Debate: Who Participates and What Do They Say?" *Journal of Urban Affairs* (2026): 1–24. https://doi.org/10.1080/07352166.2025.2601029

Luka et al. analyze 147 written depositions submitted to Montréal's public consultation body regarding the city's 2019 draft inclusionary zoning bylaw, which proposed requiring forty percent of new units at below-market rates. The qualitative content analysis reveals that seventy-nine percent of submissions supported the bylaw, with non-profit housing advocates and community organizations most represented. The study provides a qualitative policy process counterpart to the quantitative analysis planned for Philadelphia: understanding who participates in IZ policy formation and what arguments they advance contextualizes the political economy behind the MIN overlay's design. The content analysis methodology, while distinct from the permit-based approach in the capstone, offers a framework for interpreting stakeholder perspectives that could inform future qualitative case study research recommended in the trends document. The findings are specific to Montréal's political and housing context, where provincial authority over municipalities and a distinct rental market structure limit direct transferability to Philadelphia.

Mukhija, Vinit, Lara Regus, Sara Slovin, and Ashok Das. "Can Inclusionary Zoning Be an Effective and Efficient Housing Policy? Evidence from Los Angeles and Orange Counties." *Journal of Urban Affairs* 32, no. 2 (May 2010): 229–252. https://doi.org/10.1111/j.1467-9906.2010.00495.x

Mukhija et al. evaluate seventeen IZ programs adopted over thirty-five years across Los Angeles and Orange Counties, comparing their structure, affordable housing productivity, and market effects. Using cross-sectional data with unemployment rates as a proxy for market conditions, the study finds that mandatory programs are effective at producing affordable units relative to other housing programs and identifies no statistically significant adverse effect on housing supply. The authors conclude that critics overestimate IZ's supply costs while underestimating its production capacity. This directly informs the Philadelphia capstone by establishing that mandatory programs, like the MIN overlay, can be productive, provided they include cost offsets. The cross-jurisdictional comparative design, while less rigorous than the planned DiD, offers a model for evaluating policy variation across neighborhoods within the MIN. Findings are most valid for California's high-cost markets with mature IZ programs and established developer familiarity; Philadelphia's newer program and different market conditions require cautious extrapolation.

Ortiz-Villavicencio, Marcelo, Gonzalo E. Sánchez, and Mario A. Fernández. "Heterogenous Treatment Effects of a Voluntary Inclusionary Zoning Program on Housing Prices." *Housing Studies* 40, no. 3 (2025): 722–747. https://doi.org/10.1080/02673037.2024.2313592

Ortiz-Villavicencio, Sánchez, and Fernández study Auckland, New Zealand's Special Housing Areas program, a voluntary IZ initiative, using a difference-in-differences framework with quantile treatment effects estimated from approximately 175,000 sales transactions between 2011 and 2016. They find that the program failed to reduce housing prices overall, though new dwellings saw decreased prices at the lower end of the distribution and increased prices at the upper end. This heterogeneous treatment effect methodology is highly relevant to the planned DiD for Philadelphia, suggesting that average treatment effects may mask important distributional variation. The quantile approach could reveal whether the MIN overlay differentially affects smaller versus larger projects, paralleling the threshold avoidance patterns in the qualitative analysis. As a study of a voluntary program in a distinct international housing market, the findings are most applicable to contexts evaluating voluntary IZ mechanisms like Philadelphia's Mixed Income Housing Bonus, rather than the mandatory MIN overlay itself.

Schuetz, Jenny, Rachel Meltzer, and Vicki Been. "Silver Bullet or Trojan Horse? The Effects of Inclusionary Zoning on Local Housing Markets in the United States." *Urban Studies* 48, no. 2 (February 2011): 297–329. https://doi.org/10.1177/0042098009360683.

Schuetz, Meltzer, and Been use panel data from the San Francisco Bay Area and suburban Boston to estimate how inclusionary zoning affects housing prices and production, finding regionally contingent results. In suburban Boston, IZ contributed to higher prices and lower production during periods of regional appreciation, while in the Bay Area, IZ raised prices in hot markets but depressed them in cooler periods, with no statistically significant effect on new construction. The study's central finding that IZ market effects depend heavily on local economic conditions is directly applicable to this analysis, where the MIN overlay was enacted during a period of simultaneous interest rate hikes and tax abatement redesign. The panel data methodology and regional comparison framework inform the planned DiD design by demonstrating the importance of controlling for macroeconomic cycles when isolating policy-specific effects. The authors also find that total affordable unit production under IZ has been modest and depends primarily on program age, tempering expectations for the MIN's early years. Findings are most valid for the two study regions during the 1990s–2000s housing cycle and may not generalize to post-pandemic markets or cities with different regulatory structures like Philadelphia.

Wang, Ruoniu, and Xinyu Fu. "Examining the Effects of Policy Design on Affordable Unit Production Under Inclusionary Zoning Policies." *Journal of the American Planning Association* 88, no. 4 (2022): 550–564. https://doi.org/10.1080/01944363.2022.2027263

Wang and Fu analyze the relationship between IZ policy design features and affordable unit production using the Inclusionary Housing Program Database, which covers 539 IZ policies across 27 states and the District of Columbia. They find that mandatory

programs, older programs, jurisdiction-wide programs, and those with complex income requirements targeting lower-income levels produce significantly more affordable units. This large-scale cross-sectional analysis provides the broadest empirical context for evaluating the MIN overlay's design: Philadelphia's mandatory, geographically targeted approach with a forty percent AMI threshold aligns with several of the high-production design features identified. The study complements the qualitative trends document by suggesting which design elements matter most for unit production. Its findings are valid for U.S. jurisdictions with established IZ programs but may overstate expected production for newly enacted policies like the MIN, where developer learning and market adjustment are still underway.

Wang, Ruoniu (Vince), Wei Kang, and Xinyu Fu. "Do Inclusionary Zoning Policies Affect Local Housing Markets? An Empirical Study in the United States." *Cities* 158 (March 2025): 105736. https://doi.org/10.1016/j.cities.2025.105736

Wang, Kang, and Fu employ a staggered difference-in-differences design using the Grounded Solutions Network's nationwide IZ database to estimate the causal effects of IZ adoption on housing permits, rents, and home prices across U.S. municipalities. They find no average effect on permits or rents but a 2.1 percent increase in home prices, with more stringent mandatory and jurisdiction-wide policies showing larger price effects. This is the most methodologically aligned study to the planned Philadelphia DiD analysis: the staggered DiD approach accounts for heterogeneous treatment timing, directly relevant to the MIN's January 2022 enactment and subsequent boundary expansions. The null finding on permits echoes Hamilton (2021) and suggests the capstone's DiD should examine project composition and unit counts as alternative dependent variables. The study's national scope provides broad external validity, though local confounders specific to Philadelphia, particularly the concurrent tax abatement redesign, are not captured in the national model.